



tcp

terminal de
contêineres
de paranaguá



CMPort
招商港口



Strategic Study on RTGs (Transtainers)

Electrification, Retrofit & 2027 Budget Projection

Technical and financial evaluation based on maintenance history and operating hours recorded from January to August 2026. Contrasts operational intensity (operating hours) and unit cost per operating hour between **Electrified Cranes (E-RTG)** versus **Diesel (Genset) Cranes**, analyzing lifecycle degradation and projecting the 2027 maintenance budget equipment by equipment.

Period: Jan to Aug / 2026

Operating Hours: 169,263 hrs (8 months)

Total Fleet: 40 Transtainers (RTG 01 to 41)

Classification: RTG Strategic Report

1. Summary: Operational Fleet & Strategic Overview

8-MONTH REALIZED R\$ 10.85M Jan to Aug / 2026 (40 RTGs)	OPERATING HOURS (8M) 169,263 hrs Average 529 hrs/month/RTG	AVG COST PER HOUR R\$ 64.09 / hr Average maintenance cost per operating hour	2027 PROJECTION (AS-IS) R\$ 17.57M +8.0% aging escalation
---	--	--	---

Technological Composition of the 40 Transtainers (Cost vs Operating Hours)

TCP's fleet consists of **40 RTG cranes** (aged 3 to 25 years). **RTGs 01 to 03 are already operating electrified** with outstanding reliability and reduced unit cost. For the remaining diesel fleet, the data highlights batch **RTG 04 to 14** as the priority candidate for replacement:

RTG Batch	Manufacturer	Year	System	Hours (8m)	Cost R\$/hr	8-Month Cost	Monthly Avg	% Fleet
RTG 01 to 03 (3 un) Maintain Operation	KoneCranes	2001 (25y)	 E-RTG	12,564 hrs	R\$ 48.50 / hr	R\$ 609,357.13	R\$ 25,389.88	5.6%
RTG 04 to 06 (3 un) Suggested Replacement	KoneCranes	2001 (25y)	 Diesel (Genset)	9,305 hrs	R\$ 86.05 / hr	R\$ 800,696.92	R\$ 33,362.37	7.4%
RTG 08 to 10 (3 un) Suggested Replacement	Kalmar	2005 (21y)	 Diesel (Genset)	9,592 hrs	R\$ 56.15 / hr	R\$ 538,547.12	R\$ 22,439.46	5.0%
RTG 11 to 14 (4 un) Suggested Replacement (Highest Cost)	KoneCranes	2008 (18y)	 Diesel (Genset)	16,100 hrs	R\$ 162.93 / hr	R\$ 2,623,218.21	R\$ 81,975.57	24.2%
RTG 15 to 20 (6 un) Electrify (Retrofit)	KoneCranes	2011 (15y)	 Diesel (Genset)	20,041 hrs	R\$ 147.16 / hr	R\$ 2,949,297.34	R\$ 61,443.69	27.2%
RTG 21 to 30 (10 un) Electrify (Retrofit)	Kalmar	2014 (12y)	 Diesel (Genset)	42,560 hrs	R\$ 46.36 / hr	R\$ 1,973,090.67	R\$ 24,663.63	18.2%
RTG 31 to 41 (11 un) New Fleet	ZMPC	2023 (3y)	 Diesel (Genset)	59,101 hrs	R\$ 22.90 / hr	R\$ 1,353,471.05	R\$ 15,380.35	12.5%
TOTAL OPERATIONAL FLEET (40 RTGs)				169,263 hrs	R\$ 64.09 / hr	R\$ 10,847,678.44	R\$ 33,898.99	100.0%

2. Empirical Terminal Case: The Success of the E-RTG (RTG 01 to 03)

TCP features a definitive in-terminal A/B comparison: cranes from the **same manufacturer (KoneCranes)**, from the **same manufacturing year (2001)**, and with the exact same operational age (25 years):

 RTG 01 to 03 (Electrified - E-RTG) â€¢ Maintain Operation Operating Hours (8m): 12,564 hrs (Average: 524 hrs/month per crane). Cost per Operating Hour: R\$ 48.50 / hour Total Cost (8m): R\$ 609,357.13 (Average: R\$ 25,389.88/month). Availability: Delivered +35% more operating hours than the corresponding diesel model.	 RTG 04 to 06 (Diesel - Genset) â€¢ Suggested Replacement Operating Hours (8m): 9,305 hrs (Average: 388 hrs/month per crane). Cost per Operating Hour: R\$ 86.05 / hour Unit Difference: +77.4% MORE EXPENSIVE PER HOUR than electric! Limitation: Lower operating utilization resulting from frequent engine/generator breakdowns.
---	--

Proven Technical Conclusion: Because **RTGs 01 to 03 are already electrified** and operate with high efficiency (R\$ 48.50/hr), they are **maintained in service, outside the need for immediate replacement**. Priority replacement must focus on the diesel units in batch **RTG 04 to 14**.

3. Critical Analysis: KoneCranes Cost Drivers vs New ZPMC Fleet

Cross-referencing costs and operating horometers reveals that the fleet's financial bottleneck lies in the diesel KoneCranes batches. In particular, cranes **RTG 11 to 14** exhibit the highest unit cost across the entire fleet (**R\$ 162.93 / hour**), consuming **R\$ 2.62M in 8 months** despite logging significantly fewer hours than newer machines:

KONECRANES 2008 (RTG 11-14) R\$ 162.93 / hr 7.1x more expensive per hour than new ZPMC	KONECRANES 2011 (RTG 15-20) R\$ 147.16 / hr 6.4x more expensive per hour than new ZPMC	ZMPC 2023 (RTG 31-41) R\$ 22.90 / hr Peak utilization benchmark (672 hrs/month)
--	--	---

Why are RTGs 04 to 14 the priority focus for replacement?

- RTGs 04 to 06 (2001 até 25 years) & RTGs 08 to 10 (2005 até 21 years):** Diesel equipment that has reached structural design life limit. They show severe accumulated mechanical fatigue under coastal maritime exposure and critical shortages of OEM replacement parts.
- RTGs 11 to 14 (2008 até 18 years):** Represent the fleet's cost peak (R\$ 162.93/hr). The diesel powertrain suffers successive breakdowns that render further heavy corrective repairs uneconomical.
- Impact of Batch RTG 04 to 14:** Together, these 10 cranes consumed **R\$ 3,962,462.25 (36.5% of total fleet budget)**. Replacing them immediately eliminates the terminal's primary maintenance budget drain.

4. Strategic Guidelines: Fleet Electrification & Suggested Replacement (RTG 04 to 14)

Strategic Pillar	Fleet Electrification (In Progress) Focus: Mid-Life Fleet (RTG 15 to 30 até 12 to 15 years)	Replacement with New Electric RTGs Suggested Focus: Batch RTG 04 to 14 (10 critical units)
Asset Scope	RTGs 15 to 20 (KoneCranes 2011) and RTGs 21 to 30 (Kalmar 2014) with structural integrity suitable for life extension.	RTGs 04 to 14 (10 units): RTGs 04-06 (25y), RTGs 08-10 (21y), and RTGs 11-14 (18y at R\$ 162.93/hr).
RTG 01-03 Status	⚡ RTG 01 to 03 (3 units): Already successfully electrified (R\$ 48.50/hr). Remain in continuous operation, <u>excluded from replacement</u> .	
Metal Fatigue / Structural Life	Feasible: 12-to-15-year structures maintain rigidity for another 8 to 10 years of clean electric operation.	Critical / Exhausted: 18-to-25-year machines suffer accumulated microcracks, bogie wear, and severe corrosion.
Spare Parts Availability	Standard commercial busbars and inverters readily available in the market.	High Obsolescence: Original mechanical parts, gearboxes, and control boards discontinued by OEMs.

Recommended Guidelines for TCP Management:

- Maintain RTG 01 to 03 in Operation:** Since they already operate electrified with high availability (+35% hours) and a low cost of R\$ 48.50/hr, they remain active without need for immediate replacement.
- Suggested Replacement for RTG 04 to 14 (10 Cranes):** Prioritize CapEx approval for acquiring new electric yard gantries to replace RTGs 04 to 06 (25y), 08 to 10 (21y), and 11 to 14 (R\$ 162/hr), eliminating 36.5% of total fleet maintenance expenditure.
- Ongoing Electrification (RTG 15 to 41):** Advance the E-RTG conversion as rapidly as possible for the 26 cranes aged 3 to 15 years to achieve extended service life with low operating costs.

5. 2027 Budget Projection by Technological Group

Consolidated maintenance budget projection for parts and materials for 2027, integrating historical spend and operational horometer data (+10% for cranes >20y; +8% for mid-life 10 to 20y; +5% for new machines <5y):

8-MONTH REALIZED (2026) R\$ 10,847,678.44 January to August 2026	ANNUALIZED 2026 (12M) R\$ 16,271,517.66 Estimated 2026 year-end close	ESTIMATED BUDGET 2027 R\$ 17,570,790.91 "As-Is" maintenance scenario	TARGET W/ ELECTRIFICATION R\$ 12,290,000.00 Annual savings of ~R\$ 5.28M
--	---	--	--

Consolidated Matrix by Technological Batch (Financials & Operating Hours)

RTG Batch	Manufacturer	Year	System	Hours 8m	Cost R\$/hr	Realiz. 8m	Monthly Avg	Annual 2026	Escal.	Estimated 2027
RTG 01 to 03 (3 un) Maintain Operation	KoneCranes	2001 (25y)	 E-RTG	12,564 hrs	R\$ 48.50 / hr	R\$ 609,357.13	R\$ 76,169.64	R\$ 914,035.70	+10%	R\$ 1,005,439.26
RTG 04 to 06 (3 un) Suggested Replacement	KoneCranes	2001 (25y)	 Diesel (Genset)	9,305 hrs	R\$ 86.05 / hr	R\$ 800,696.92	R\$ 100,087.12	R\$ 1,201,045.38	+10%	R\$ 1,321,149.92
RTG 08 to 10 (3 un) Suggested Replacement	Kalmar	2005 (21y)	 Diesel (Genset)	9,592 hrs	R\$ 56.15 / hr	R\$ 538,547.12	R\$ 67,318.39	R\$ 807,820.68	+10%	R\$ 888,602.75
RTG 11 to 14 (4 un) Suggested Replacement (Highest Cost)	KoneCranes	2008 (18y)	 Diesel (Genset)	16,100 hrs	R\$ 162.93 / hr	R\$ 2,623,218.21	R\$ 327,902.28	R\$ 3,934,827.32	+8%	R\$ 4,249,613.50
RTG 15 to 20 (6 un) Electrify (Retrofit)	KoneCranes	2011 (15y)	 Diesel (Genset)	20,041 hrs	R\$ 147.16 / hr	R\$ 2,949,297.34	R\$ 368,662.17	R\$ 4,423,946.01	+8%	R\$ 4,777,861.69
RTG 21 to 30 (10 un) Electrify (Retrofit)	Kalmar	2014 (12y)	 Diesel (Genset)	42,560 hrs	R\$ 46.36 / hr	R\$ 1,973,090.67	R\$ 246,636.33	R\$ 3,127,709.24	+8%	R\$ 3,377,925.97
RTG 31 to 41 (11 un) New Fleet	ZMPC	2023 (3y)	 Diesel (Genset)	59,101 hrs	R\$ 22.90 / hr	R\$ 1,353,471.05	R\$ 169,183.88	R\$ 2,029,743.89	+5%	R\$ 2,131,230.81
FLEET CONSOLIDATED TOTAL (40 RTGs)				169,263 hrs	R\$ 64.09 / hr	R\$ 10,847,678.44	R\$ 1,355,959.81	R\$ 16,271,517.66	+8.0%	R\$ 17,570,790.91

Detailed Methodology & Calculation Premises

- Hours 8m:** Total cumulative hours recorded on each RTG's horometer from January to August 2026.
- Cost R\$/hr:** Direct maintenance material spend divided by operating hours (Realiz. 8m / Hours 8m).
- Realiz. 8m:** Total material requisitions and spare parts consumed from 01/01 to 08/31/2026 (Base V2).
- Monthly Avg:** 8-month realized spend divided by exactly 8 operating months.
- Annual 2026:** Linear 12-month run rate (Monthly Avg * 12 or Realiz. 8m * 1.5).
- Estimated 2027:** Projected maintenance budget for 2027 (Annual 2026 * (1 + Escal/100)).

* Individual equipment breakdown for each of the 40 RTGs detailing operating hours, hourly cost, and budget projection is presented on the following pages.

6. Individual Fleet Breakdown (Part 1: RTG #01 to #20)

Individual maintenance costs and operating hours for the first 20 Transtainers (highest budget impact):

#	Equipment	Manufacturer	Year	Hours 8m	Cost R\$/hr	Realiz. 8m	Monthly Avg	Annual 2026	Escal.	Estimated 2027
#1	RTG 11 (Tag 711)	KoneCranes	2008 (18y)	4,174 hrs	R\$ 194.72	R\$ 812,764.19	R\$ 101,595.52	R\$ 1,219,146.29	+8%	R\$ 1,316,677.99
#2	RTG 18 (Tag 718)	KoneCranes	2011 (15y)	3,241 hrs	R\$ 237.46	R\$ 769,620.22	R\$ 96,202.53	R\$ 1,154,430.33	+8%	R\$ 1,246,784.76
#3	RTG 12 (Tag 712)	KoneCranes	2008 (18y)	4,048 hrs	R\$ 185.64	R\$ 751,472.60	R\$ 93,934.08	R\$ 1,127,208.90	+8%	R\$ 1,217,385.61
#4	RTG 13 (Tag 713)	KoneCranes	2008 (18y)	4,342 hrs	R\$ 157.10	R\$ 682,109.46	R\$ 85,263.68	R\$ 1,023,164.19	+8%	R\$ 1,105,017.33
#5	RTG 17 (Tag 717)	KoneCranes	2011 (15y)	3,203 hrs	R\$ 190.62	R\$ 610,542.61	R\$ 76,317.83	R\$ 915,813.92	+8%	R\$ 989,079.03
#6	RTG 15 (Tag 715)	KoneCranes	2011 (15y)	4,737 hrs	R\$ 100.32	R\$ 475,204.90	R\$ 59,400.61	R\$ 712,807.35	+8%	R\$ 769,831.94
#7	RTG 16 (Tag 716)	KoneCranes	2011 (15y)	4,313 hrs	R\$ 108.18	R\$ 466,594.17	R\$ 58,324.27	R\$ 699,891.26	+8%	R\$ 755,882.56
#8	RTG 19 (Tag 719)	KoneCranes	2011 (15y)	4,547 hrs	R\$ 99.58	R\$ 452,801.73	R\$ 56,600.22	R\$ 679,202.60	+8%	R\$ 733,538.80
#9	RTG 38 (Tag 738)	ZMPC	2023 (3y)	5,716 hrs	R\$ 68.04	R\$ 388,940.53	R\$ 48,617.57	R\$ 583,410.80	+5%	R\$ 612,581.33
#10	RTG 14 (Tag 714)	KoneCranes	2008 (18y)	3,536 hrs	R\$ 106.58	R\$ 376,871.96	R\$ 47,109.00	R\$ 565,307.94	+8%	R\$ 610,532.58
#11	RTG 01 (Tag 701)	KoneCranes	2001 (25y)	3,305 hrs	R\$ 106.99	R\$ 353,591.03	R\$ 44,198.88	R\$ 530,386.55	+10%	R\$ 583,425.20
#12	RTG 06 (Tag 706)	KoneCranes	2001 (25y)	3,807 hrs	R\$ 82.01	R\$ 312,225.70	R\$ 39,028.21	R\$ 468,338.55	+10%	R\$ 515,172.41
#13	RTG 09 (Tag 709)	Kalmar	2005 (21y)	3,937 hrs	R\$ 77.92	R\$ 306,784.26	R\$ 38,348.03	R\$ 460,176.39	+10%	R\$ 506,194.03
#14	RTG 21 (Tag 721)	Kalmar	2014 (12y)	4,285 hrs	R\$ 72.28	R\$ 309,711.03	R\$ 38,713.88	R\$ 464,566.55	+8%	R\$ 501,731.87
#15	RTG 25 (Tag 725)	Kalmar	2014 (12y)	4,087 hrs	R\$ 73.59	R\$ 300,756.14	R\$ 37,594.52	R\$ 451,134.21	+8%	R\$ 487,224.95
#16	RTG 23 (Tag 723)	Kalmar	2014 (12y)	4,335 hrs	R\$ 63.47	R\$ 275,136.40	R\$ 34,392.05	R\$ 412,704.60	+8%	R\$ 445,720.97
#17	RTG 05 (Tag 705)	KoneCranes	2001 (25y)	1,975 hrs	R\$ 124.36	R\$ 245,616.80	R\$ 30,702.10	R\$ 368,425.20	+10%	R\$ 405,267.72
#18	RTG 04 (Tag 704)	KoneCranes	2001 (25y)	3,523 hrs	R\$ 68.93	R\$ 242,854.42	R\$ 30,356.80	R\$ 364,281.63	+10%	R\$ 400,709.79
#19	RTG 24 (Tag 724)	Kalmar	2014 (12y)	4,793 hrs	R\$ 43.05	R\$ 206,328.88	R\$ 25,791.11	R\$ 309,493.32	+8%	R\$ 334,252.79
#20	RTG 22 (Tag 722)	Kalmar	2014 (12y)	4,933 hrs	R\$ 40.03	R\$ 197,484.31	R\$ 24,685.54	R\$ 296,226.47	+8%	R\$ 319,924.58
SUBTOTAL CRITICAL GROUP (RTG #01 to #20)				80,837 hrs	R\$ 105.61	R\$ 8,537,411.34	R\$ 1,067,176.42	R\$ 12,806,117.01	+8.1%	R\$ 13,856,936.21

* Continued with the remaining 20 RTGs (#21 to #40) and consolidated fleet total on Page 6.

6. Individual Fleet Breakdown (Part 2: RTG #21 to #40 & Grand Total)

Individual maintenance costs and operating hours for Transtainers #21 to #40 and fleet total consolidation:

#	Equipment	Manufacturer	Year	Hours 8m	Cost R\$/hr	Realiz. 8m	Monthly Avg	Annual 2026	Escal.	Estimated 2027
#21	RTG 27 (Tag 727)	Kalmar	2014 (12y)	4,784 hrs	R\$ 40.11	R\$ 191,907.73	R\$ 23,988.47	R\$ 287,861.60	+8%	R\$ 310,890.52
#22	RTG 20 (Tag 720)	KoneCranes	2011 (15y)	0 hrs (idle)	â€”	R\$ 174,533.71	R\$ 21,816.71	R\$ 261,800.57	+8%	R\$ 282,744.61
#23	RTG 10 (Tag 710)	Kalmar	2005 (21y)	4,325 hrs	R\$ 37.86	R\$ 163,752.08	R\$ 20,469.01	R\$ 245,628.12	+10%	R\$ 270,190.93
#24	RTG 28 (Tag 728)	Kalmar	2014 (12y)	5,002 hrs	R\$ 31.61	R\$ 158,096.81	R\$ 19,762.10	R\$ 237,145.22	+8%	R\$ 256,116.83
#25	RTG 26 (Tag 726)	Kalmar	2014 (12y)	4,889 hrs	R\$ 30.93	R\$ 151,240.83	R\$ 18,905.10	R\$ 226,861.25	+8%	R\$ 245,010.14
#26	RTG 41 (Tag 741)	ZMPC	2023 (3y)	4,735 hrs	R\$ 31.78	R\$ 150,486.97	R\$ 18,810.87	R\$ 225,730.46	+5%	R\$ 237,016.98
#27	RTG 03 (Tag 703)	KoneCranes	2001 (25y)	3,931 hrs	R\$ 33.65	R\$ 132,262.24	R\$ 16,532.78	R\$ 198,393.36	+10%	R\$ 218,232.70
#28	RTG 37 (Tag 737)	ZMPC	2023 (3y)	5,739 hrs	R\$ 23.50	R\$ 134,840.43	R\$ 16,855.05	R\$ 202,260.65	+5%	R\$ 212,373.68
#29	RTG 30 (Tag 730)	Kalmar	2014 (12y)	5,452 hrs	R\$ 23.88	R\$ 130,196.30	R\$ 16,274.54	R\$ 195,294.45	+8%	R\$ 210,918.01
#30	RTG 02 (Tag 702)	KoneCranes	2001 (25y)	5,328 hrs	R\$ 23.18	R\$ 123,503.86	R\$ 15,437.98	R\$ 185,255.79	+10%	R\$ 203,781.37
#31	RTG 31 (Tag 731)	ZMPC	2023 (3y)	5,320 hrs	R\$ 20.73	R\$ 110,301.56	R\$ 13,787.70	R\$ 165,452.34	+5%	R\$ 173,724.96
#32	RTG 36 (Tag 736)	ZMPC	2023 (3y)	5,721 hrs	R\$ 17.00	R\$ 97,251.31	R\$ 12,156.41	R\$ 145,876.97	+5%	R\$ 153,170.81
#33	RTG 35 (Tag 735)	ZMPC	2023 (3y)	5,723 hrs	R\$ 14.78	R\$ 84,582.01	R\$ 10,572.75	R\$ 126,873.02	+5%	R\$ 133,216.67
#34	RTG 40 (Tag 740)	ZMPC	2023 (3y)	4,547 hrs	R\$ 18.14	R\$ 82,504.89	R\$ 10,313.11	R\$ 123,757.34	+5%	R\$ 129,945.20
#35	RTG 32 (Tag 732)	ZMPC	2023 (3y)	5,581 hrs	R\$ 14.40	R\$ 80,357.71	R\$ 10,044.71	R\$ 120,536.57	+5%	R\$ 126,563.39
#36	RTG 34 (Tag 734)	ZMPC	2023 (3y)	4,565 hrs	R\$ 17.43	R\$ 79,575.21	R\$ 9,946.90	R\$ 119,362.82	+5%	R\$ 125,330.96
#37	RTG 33 (Tag 733)	ZMPC	2023 (3y)	5,765 hrs	R\$ 12.62	R\$ 72,758.19	R\$ 9,094.77	R\$ 109,137.29	+5%	R\$ 114,594.15
#38	RTG 39 (Tag 739)	ZMPC	2023 (3y)	5,689 hrs	R\$ 12.63	R\$ 71,872.24	R\$ 8,984.03	R\$ 107,808.36	+5%	R\$ 113,198.78
#39	RTG 08 (Tag 708)	Kalmar	2005 (21y)	1,330 hrs	R\$ 51.14	R\$ 68,010.78	R\$ 8,501.35	R\$ 102,016.17	+10%	R\$ 112,217.79
#40	RTG 29 (Tag 729)	Kalmar	2014 (12y)	0 hrs (idle)	â€”	R\$ 52,232.24	R\$ 6,529.03	R\$ 78,348.36	+8%	R\$ 84,616.23
FLEET GRAND TOTAL (40 RTGs)				169,263 hrs	R\$ 64.09	R\$ 10,847,678.44	R\$ 1,355,959.81	R\$ 16,271,517.66	+8.0%	R\$ 17,570,790.91

Methodological Note: Figures refer strictly to direct crane maintenance parts and consumable materials. Excludes external third-party service contracts (R\$ 81k in 2026) and facility/workshop W900 administrative costs (R\$ 9k). Cumulative operating hours extracted from the official Jan-Aug 2026 terminal horometer report.